

APPENDIX B

FAULT TREE ANALYSIS (FTA)

Drone-Remote Weapon Station
MIL-STD-882E System Safety Assessment Report Appendix

Document purpose	Provide qualitative FTA trees, hazard traceability, mitigations, and verification evidence in SAR-appendix form.
Source basis	MIL-STD-882E System Safety document, including PHA list, 40-hazard SHA log, verification procedures, and architecture notes.
FTA scope	One tree for each PHA hazard category
Method	Top-down qualitative fault tree analysis with OR / AND logic, transfer nodes, hazard-ID traceability, and mitigation / verification linkage.
Architecture emphasis	Explicit treatment of DRONE Flight Control <-> Turret Control <-> Recoil Mitigation couplings, plus power, sensor, comms, and retention paths.

1. Purpose and Scope

- This appendix presents qualitative Fault Tree Analysis (FTA) artifacts for the drone-mounted remote weapon station in a format suitable for a MIL-STD-882E style Safety Assessment Report (SAR).
- The appendix supplements the uploaded System Hazard Analysis by decomposing each major Preliminary Hazard Analysis (PHA) item into top events, intermediate events, and transferable basic-event groupings tied directly to the hazard log.

2. Source Material Used

- The appendix was built from the source document sections covering the PHA list, the 40-hazard preliminary RAC table, the fail-safe CONOPS, the safety verification matrix, the SHA interaction diagram, the recoil-flight coupling analysis, the FMEA detection rationale, the cyber-safety interface, HFE / EMI requirements, and the fire-authorization chain.

3. FTA Conventions

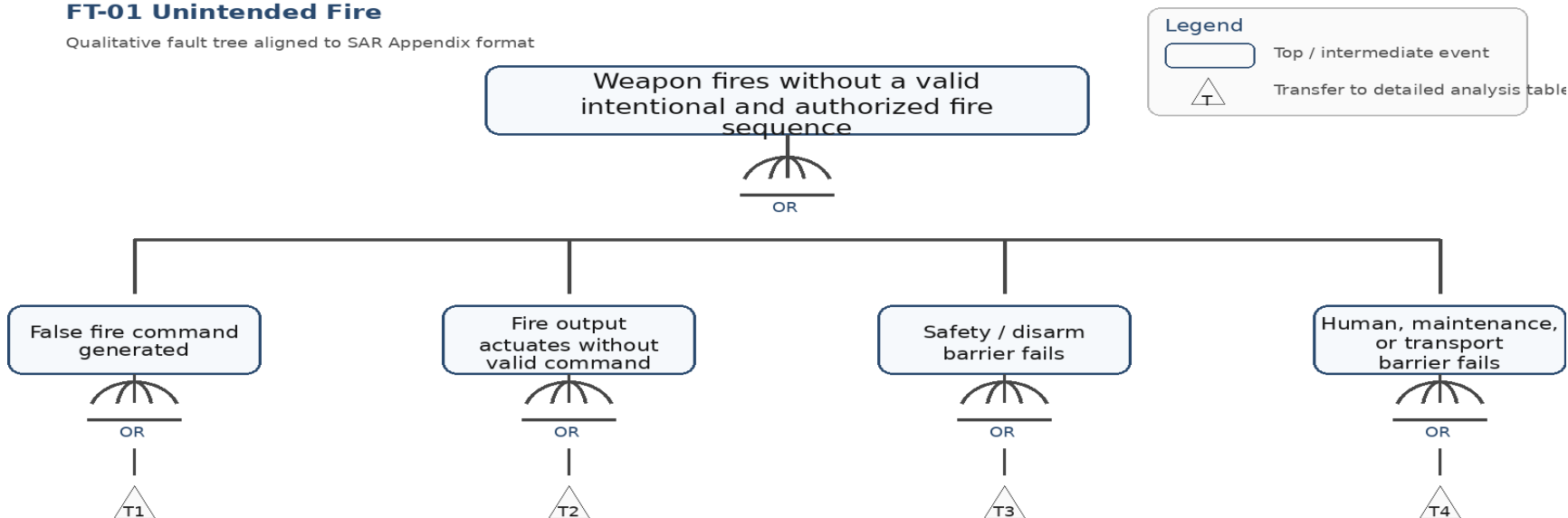
Symbol / construct	Meaning	Use in this appendix
Top / intermediate event box	Hazardous state or logical contributor	Shown directly on each tree
AND gate	All conditions required	Used for combined failure conditions such as loss of link AND failed auto-DISARM
OR gate	Any condition sufficient	Used for alternative cause branches
Transfer node	Reference to detailed branch analysis table	Used to keep the figure readable while preserving traceability

4. FT-01 Unintended Fire

PHA hazard	Unintended fire	Related SHA hazard IDs	H-1, H-2, H-3, H-4, H-16, H-17, H-32, H-33, H-34, H-37
Top event	Weapon fires without a valid intentional and authorized fire sequence		
Tree logic	OR	Primary appendix figure	Figure FT-01
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	This tree supports the highest-severity unauthorized-fire hazards in the hazard log. It shows that the design relies on layered prevention: command plausibility, independent hardware inhibit, and procedural barriers.		

FT-01 Unintended Fire

Qualitative fault tree aligned to SAR Appendix format



FT-01 Unintended Fire

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-1, H-32	SW logic fault; mode-handling error; inadvertent mode change	H-1: Fire-control SW defect, mode-handling error H-32: UI design; accidental click	H-1: Independent ARM state machine; dual-confirm (operator + guarded ARM); two-channel FIRE command ... H-32: Guarded mode switch; confirm dialog; logging; audible/visual annunciation	H-1: SW unit/integration tests; fault-injection; HIL bench test; H-32: Human factors test; UI validation
T2	H-2, H-16, H-37	Driver/relay stuck ON; trigger linkage or solenoid stuck; shot feedback hidden	H-2: Relay/MOSFET short, driver failure H-16: Mechanical trigger linkage jam; solenoid stuck H-37: Sensor disconnected	H-2: Series hardware inhibit (safety relay + mechanical sear/solenoid); monitored output feedback; H-16: Burst limiter; cut-off timer; redundant inhibit; monitored firing feedback (shot counter) H-37: Independent burst limiter; redundant detection (acoustic/accel optional); health monitoring	H-2: Bench electrical fault tests (short/open); output feedback validation; firing inhibit H-16: Burst-limit test; stuck-solenoid fault test; live-fire demo H-37: Sensor disconnect tests; burst limiter demo
T3	H-17, H-2, H-1	Failure to DISARM; safety relay/interlock stuck; BIT/watchdog ineffective	H-17: SW state latch; hardware interlock stuck H-2: Relay/MOSFET short, driver failure H-1: Fire-control SW defect, mode-handling error	H-17: Disarm in hardware + software; independent safety relay; disarm-on-fault; BIT; maintenance lock... H-2: Series hardware inhibit (safety relay + mechanical sear/solenoid); monitored output feedback; c... H-1: Independent ARM state machine; dual-confirm (operator + guarded ARM); two-channel FIRE command ...	H-17: Fault-injection; bench tests; safety relay verification H-2: Bench electrical fault tests (short/open); output feedback validation; firing inhibit t... H-1: SW unit/integration tests; fault-injection; HIL bench test; firing test with induced fa...
T4	H-3, H-33, H-34	Operator error; maintenance arming; transport actuation	H-3: Mode confusion; accidental trigger; UI ambiguity H-33: Procedure lapse; service mode missing H-34: Vibration; packed state	H-3: Two-step FIRE (arm + confirm); guarded control; clear UI state; required positive ID & authoriz... H-33: Maintenance lockout/tagout; physical safety pin; service mode disables firing circuits; SOP H-34: Transport lock; mechanical restraint; SAFE state; covers; shipping procedure	H-3: Human factors review; UI simulation; operator eval; procedural compliance drills H-33: Maintenance procedure audit; lockout verification H-34: Transport vibration test; pack-out inspection

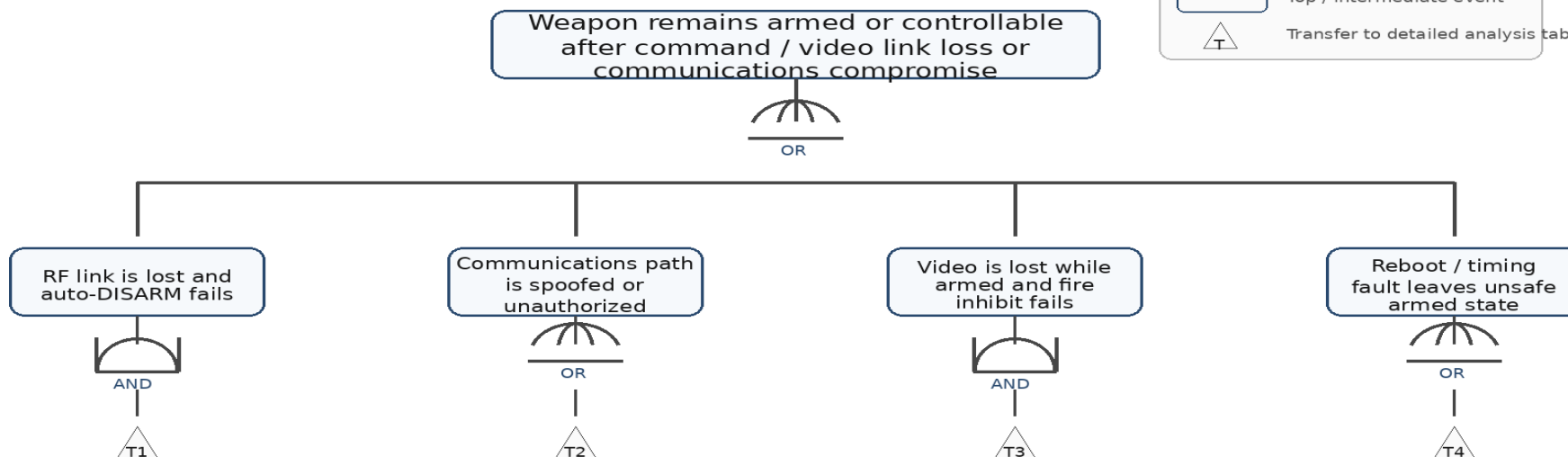
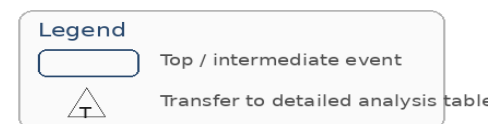
FTA assessment: This tree supports the highest-severity unauthorized-fire hazards in the hazard log. It shows that the design relies on layered prevention: command plausibility, independent hardware inhibits, and procedural barriers. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

5. FT-02 Loss of Communications While Armed

PHA hazard	Loss of comms armed	Related SHA hazard IDs	H-8, H-9, H-30, H-31, H-40
Top event	Weapon remains armed or controllable after command / video link loss or communications compromise		
Tree logic	OR	Primary appendix figure	Figure FT-02
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	The safety argument depends on deterministic transition to SAFE on loss of heartbeat or video, plus re-authentication before re-entry to ARMED state.		

FT-02 Loss of Communications While Armed

Qualitative fault tree aligned to SAR Appendix format



בלמ"ס – מסווג מסחרית

FT-02 Loss of Communications While Armed

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-8, H-40	RF link drop or jam occurs and auto-DISARM path fails	H-8: RF link drop/jam; integration fault H-40: Camera failure; link congestion	H-8: Heartbeat supervision; automatic DISARM on link loss; local timer to safe; H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm;	H-8: Comm-loss tests (RF off/jam); timing verification; flight test H-40: Video loss tests; comm congestion tests;
T2	H-9	Weak authentication; spoofed packets; key compromise	H-9: Weak auth; key compromise; spoofed packets	H-9: Strong encryption + mutual auth; rolling keys; whitelist device IDs; command signing; anti-repl...	H-9: Pen-test; protocol tests; replay/spoof tests; code review
T3	H-40, H-8	Video lost while armed and inhibit FIRE without live video fails	H-40: Camera failure; link congestion H-8: RF link drop/jam; integration fault	H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l... H-8: Heartbeat supervision; automatic DISARM on link loss; local timer to safe; operator re-auth req...	H-40: Video loss tests; comm congestion tests; inhibit demo H-8: Comm-loss tests (RF off/jam); timing verification; flight test
T4	H-30, H-31	Software crash/reboot or time sync defect leaves unsafe state	H-30: Unhandled exception; memory leak H-31: Time drift; GPS time loss	H-30: Safe-state on reboot; watchdog; state machine recovery; redundancy/backup controller H-31: Monotonic timers; sanity checks; disable time-based gating on invalid time; alarm	H-30: Stress tests; fault-injection; long-run endurance; reboot recovery demo H-31: Time-drift tests; GPS-loss tests

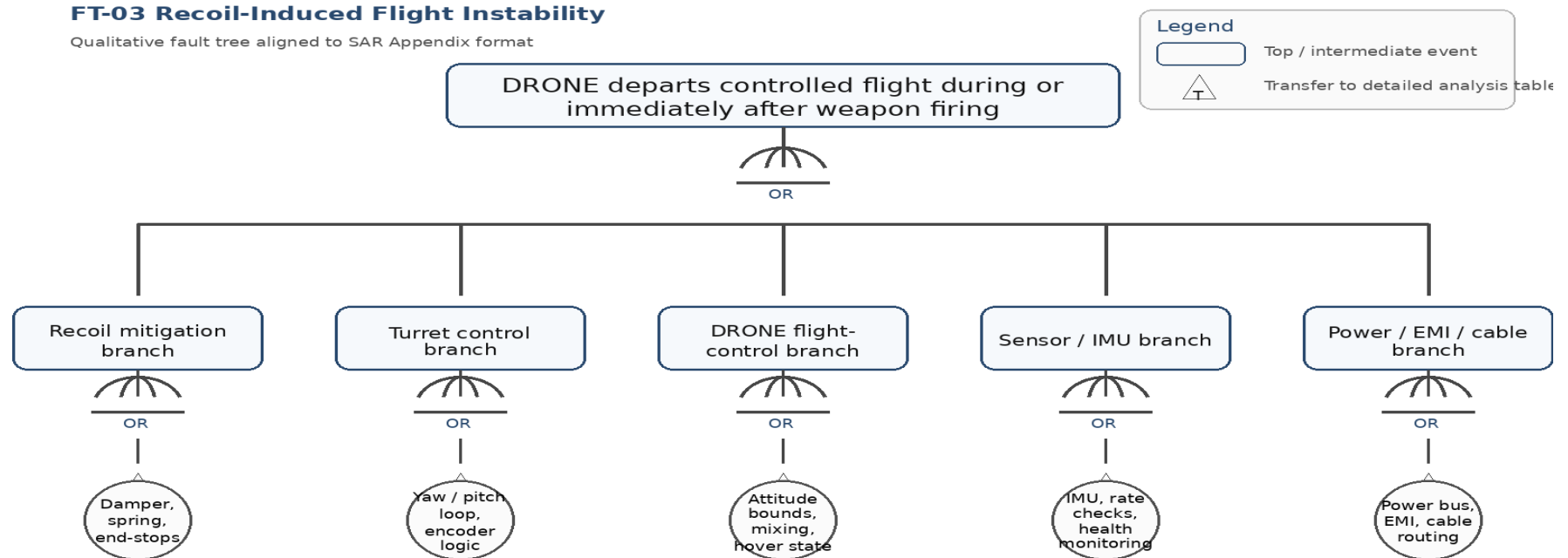
FTA assessment: The safety argument depends on deterministic transition to SAFE on loss of heartbeat or video, plus re-authentication before re-entry to ARMED state. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

6. FT-03 Recoil-Induced Flight Instability

PHA hazard	Recoil instability	Related SHA hazard IDs	H-11, H-12, H-28, H-29, H-38
Top event	DRONE departs controlled flight during or immediately after weapon firing		
Tree logic	OR	Primary appendix figure	Figure FT-03
Architecture focus	Matches SHA subsystem couplings and fail-safe barriers		
Analysis note	This architecture-specific tree matches the SHA interaction model: DRONE Flight Control <-> Turret Control <-> Recoil Mitigation, with power and sensor couplings. The critical path is recoil disturbance combined with failure of stability protection or flight-envelope interlocks.		

FT-03 Recoil-Induced Flight Instability

Qualitative fault tree aligned to SAR Appendix format



FT-03 Recoil-Induced Flight Instability

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-12	Damper leak/break; spring fracture; recoil cancellation degraded	H-12: Damper leak/break; spring fracture	H-12: Mechanical margins; end-stops; condition monitoring; scheduled inspection; inhibit sustained fi...	H-12: Endurance firing test; inspection; vibration test
T2	H-11, H-38	Turret control instability; encoder slip; unsafe burst / rate-of-fire	H-11: Stabilizer control fault; IMU failure H-38: Coupled dynamics; control mixing	H-11: Recoil mitigation; stabilization health monitoring; inhibit burst if stability out-of-bounds; r... H-38: Integration analysis; limit firing to stable hover; interlock with DRONE attitude bounds; tuning	H-11: Firing tests; IMU fault tests; stability threshold verification H-38: Flight tests; coupled-model simulation; firing in flight validation
T3	H-38	Flight-controller coupling; firing outside stable hover or attitude bounds	H-38: Coupled dynamics; control mixing	H-38: Integration analysis; limit firing to stable hover; interlock with DRONE attitude bounds; tuning	H-38: Flight tests; coupled-model simulation; firing in flight validation
T4	H-11	IMU failure/inconsistency; stabilization health monitor ineffective	H-11: Stabilizer control fault; IMU failure	H-11: Recoil mitigation; stabilization health monitoring; inhibit burst if stability out-of-bounds; r...	H-11: Firing tests; IMU fault tests; stability threshold verification
T5	H-28, H-29, H-19	Cable snag, EMI upset, or power competition perturb control loops	H-28: Routing error; wear H-29: RF emitters on DRONE; payload interference H-19: Battery drop; connector intermittent	H-28: Cable management; strain relief; clearance analysis; protective sleeving H-29: EMI shielding; filtering; separation; grounding; software robustness H-19: Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth	H-28: Fit-check; vibration test; flight envelope inspection H-29: EMI tests; on-platform integration test H-19: Power interruption tests; connector wiggle tests

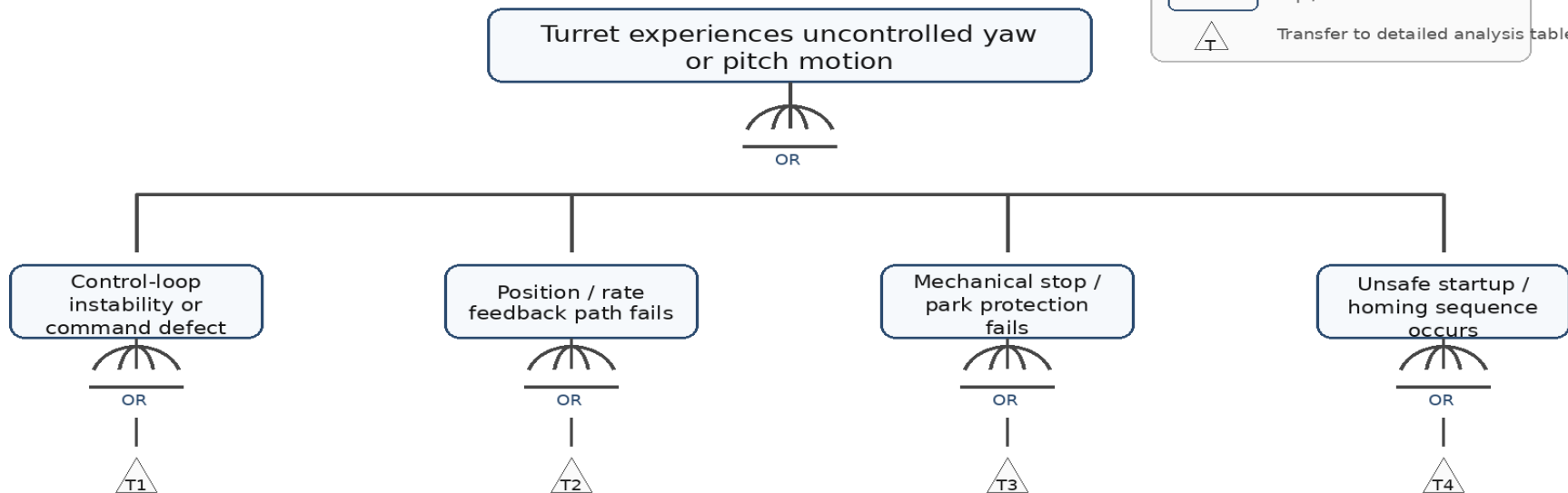
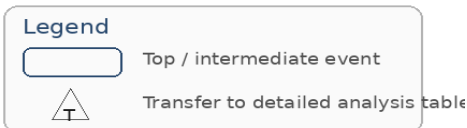
FTA assessment: This architecture-specific tree matches the SHA interaction model: DRONE Flight Control <-> Turret Control <-> Recoil Mitigation, with power and sensor couplings. The critical path is recoil disturbance combined with failure of stability protection or flight-envelope interlocks. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

7. FT-04 Turret Runaway Motion

PHA hazard	Turret runaway	Related SHA hazard IDs	H-6, H-7, H-18, H-27
Top event	Turret experiences uncontrolled yaw or pitch motion		
Tree logic	OR	Primary appendix figure	Figure FT-04
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	This tree captures both in-motion runaway and unexpected startup motion. It demonstrates the role of limiters, stop strength, and inhibit-until-homed logic.		

FT-04 Turret Runaway Motion

Qualitative fault tree aligned to SAR Appendix format



FT-04 Turret Runaway Motion

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-6, H-7	Control loop instability; bad command; rate/position limiter failure	H-6: Control loop instability; sensor fault; encoder slip H-7: Control loop instability; IMU/encoder fault	H-6: Rate/position limiters; watchdog; torque/current limiting; mechanical stops; redundant encoder ... H-7: As above + pitch hard-stop; gravity-compensation limits; safe park routine	H-6: HIL stability tests; induced sensor fault; motion envelope tests H-7: Motion tests at extremes; fault injection; safe-park demo
T2	H-6, H-7, H-18	Encoder / IMU fault; slip; plausibility check ineffective	H-6: Control loop instability; sensor fault; encoder slip H-7: Control loop instability; IMU/encoder fault H-18: No homing; encoder offset	H-6: Rate/position limiters; watchdog; torque/current limiting; mechanical stops; redundant encoder ... H-7: As above + pitch hard-stop; gravity-compensation limits; safe park routine H-18: Safe startup: power-up in SAFE; homing sequence; park pose; inhibit fire until homed; warnings	H-6: HIL stability tests; induced sensor fault; motion envelope tests H-7: Motion tests at extremes; fault injection; safe-park demo H-18: Power-cycle tests; homing validation; safe-start demo
T3	H-27, H-7	Mechanical stop wear/fracture or safe-park routine fails	H-27: Stop fracture; wear H-7: Control loop instability; IMU/encoder fault	H-27: Stop strength margin; redundant limit switches; current limiting; periodic inspection H-7: As above + pitch hard-stop; gravity-compensation limits; safe park routine	H-27: End-stop overload test; inspection; motion tests H-7: Motion tests at extremes; fault injection; safe-park demo
T4	H-18	Startup homing/park sequencing defect causes unexpected motion	H-18: No homing; encoder offset	H-18: Safe startup: power-up in SAFE; homing sequence; park pose; inhibit fire until homed; warnings	H-18: Power-cycle tests; homing validation; safe-start demo

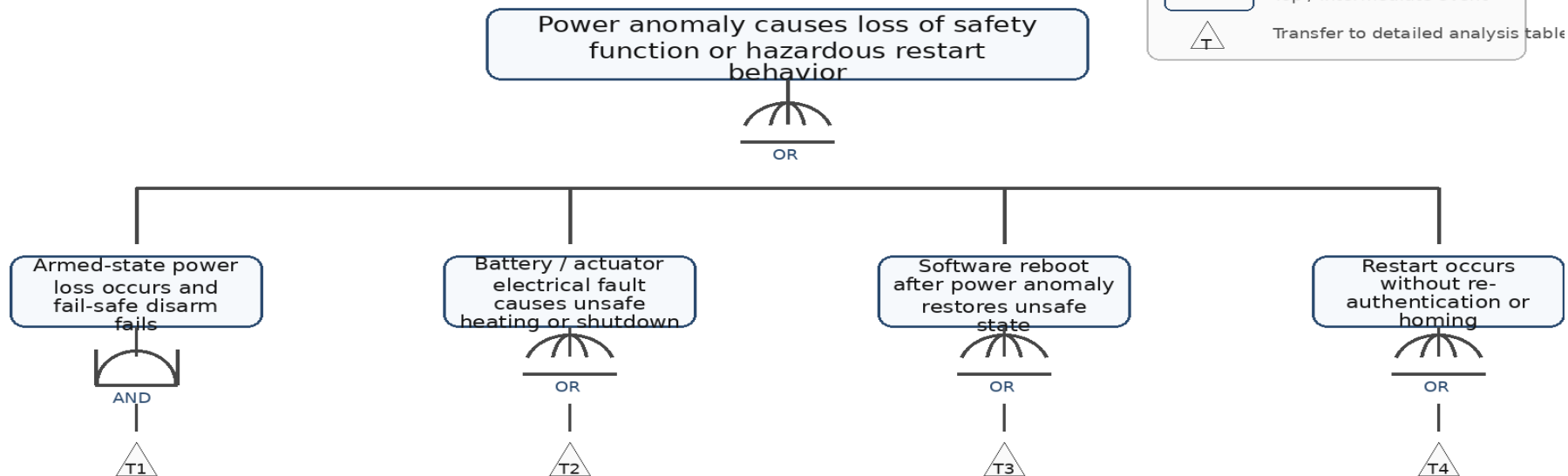
FTA assessment: This tree captures both in-motion runaway and unexpected startup motion. It demonstrates the role of limiters, stop strength, and inhibit-until-home logic. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

8. FT-05 Power Loss or Unsafe Re-Energization

PHA hazard	Power loss	Related SHA hazard IDs	H-19, H-20, H-30, H-36
Top event	Power anomaly causes loss of safety function or hazardous restart behavior		
Tree logic	OR	Primary appendix figure	Figure FT-05
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	Power anomalies are safety-significant because they can simultaneously affect command, actuation, and monitoring. The required design behavior is DISARM + re-authenticate + safe homing.		

FT-05 Power Loss or Unsafe Re-Energization

Qualitative fault tree aligned to SAR Appendix format



FT-05 Power Loss or Unsafe Re-Energization

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-19	Battery drop/connector intermittency and disarm-on-power-loss fails	H-19: Battery drop; connector intermittent	H-19: Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth	H-19: Power interruption tests; connector wiggle tests
T2	H-20, H-36	Battery thermal runaway or actuator overcurrent causes electrical upset	H-20: Battery defect; overcurrent H-36: Stall; short; overheating motor	H-20: BMS protection; fusing; thermal isolation; venting; temperature monitoring; charging controls H-36: Current limiting; fuses; temperature monitoring; stall detection and shutdown	H-20: Abuse tests (as applicable); thermal monitoring validation; inspection H-36: Stall tests; thermal monitoring validation
T3	H-30, H-19	Crash/reboot recovery restores unsafe armed state	H-30: Unhandled exception; memory leak H-19: Battery drop; connector intermittent	H-30: Safe-state on reboot; watchdog; state machine recovery; redundancy/backup controller H-19: Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth	H-30: Stress tests; fault-injection; long-run endurance; reboot recovery demo H-19: Power interruption tests; connector wiggle tests
T4	H-18, H-19, H-30	Restart bypasses re-authentication, SAFE initialization, or homing	H-18: No homing; encoder offset H-19: Battery drop; connector intermittent H-30: Unhandled exception; memory leak	H-18: Safe startup: power-up in SAFE; homing sequence; park pose; inhibit fire until homed; warnings H-19: Fail-safe to DISARM on power loss; mechanical safe; state recorded; restart requires re-auth H-30: Safe-state on reboot; watchdog; state machine recovery; redundancy/backup controller	H-18: Power-cycle tests; homing validation; safe-start demo H-19: Power interruption tests; connector wiggle tests H-30: Stress tests; fault-injection; long-run endurance; reboot recovery demo

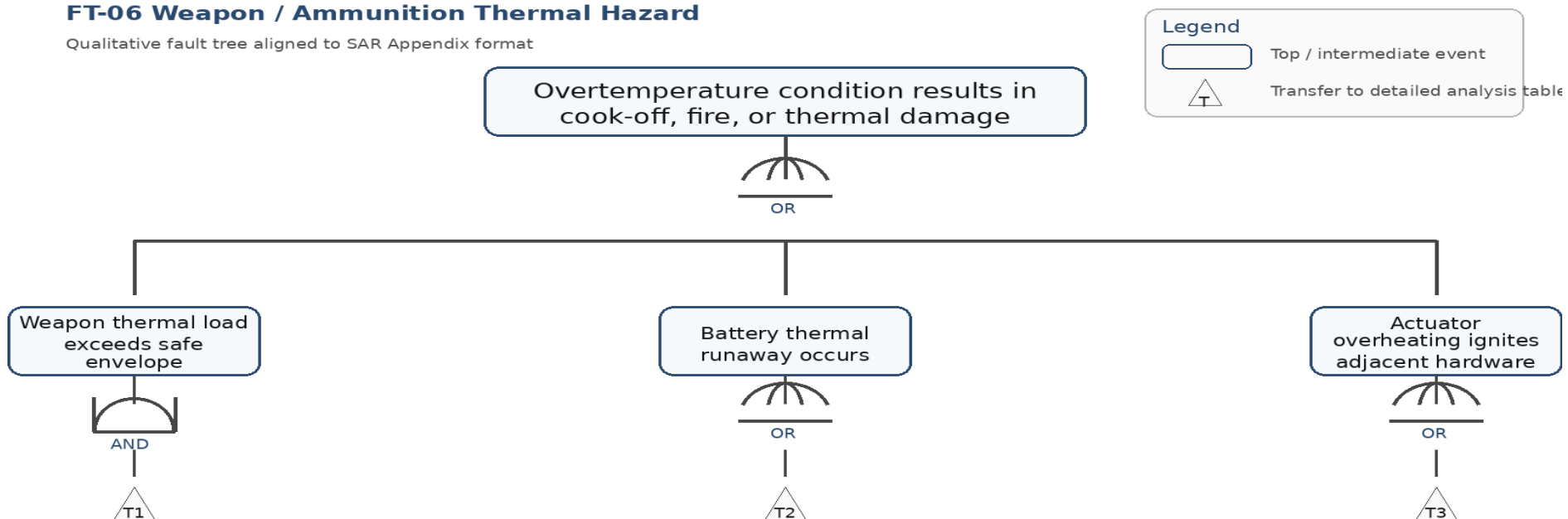
FTA assessment: Power anomalies are safety-significant because they can simultaneously affect command, actuation, and monitoring. The required design behavior is DISARM + re-authenticate + safe homing. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

9. FT-06 Weapon / Ammunition Thermal Hazard

PHA hazard	Ammo cook-off	Related SHA hazard IDs	H-14, H-20, H-36
Top event	Overtemperature condition results in cook-off, fire, or thermal damage		
Tree logic	OR	Primary appendix figure	Figure FT-06
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	This tree ties thermal hazards to duty-cycle control, barrel temperature sensing, BMS controls, and stall shutdown. The top event is prevented by both limiting heat generation and reliably inhibiting continued firing.		

FT-06 Weapon / Ammunition Thermal Hazard

Qualitative fault tree aligned to SAR Appendix format



בלמ"ס – מסווג מסחרית

FT-06 Weapon / Ammunition Thermal Hazard

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-14	High burst / blocked cooling occurs and thermal inhibit fails	H-14: High burst; blocked cooling	H-14: Thermal model; barrel temp sensor; firing duty-cycle limits; automatic cooldown inhibit; warnin...	H-14: Thermal tests; firing schedule validation; sensor calibration
T2	H-20	Battery defect / overcurrent bypasses BMS, venting, or thermal isolation	H-20: Battery defect; overcurrent	H-20: BMS protection; fusing; thermal isolation; venting; temperature monitoring; charging controls	H-20: Abuse tests (as applicable); thermal monitoring validation; inspection
T3	H-36	Motor stall/short and shutdown fails, causing secondary fire	H-36: Stall; short; overheating motor	H-36: Current limiting; fuses; temperature monitoring; stall detection and shutdown	H-36: Stall tests; thermal monitoring validation

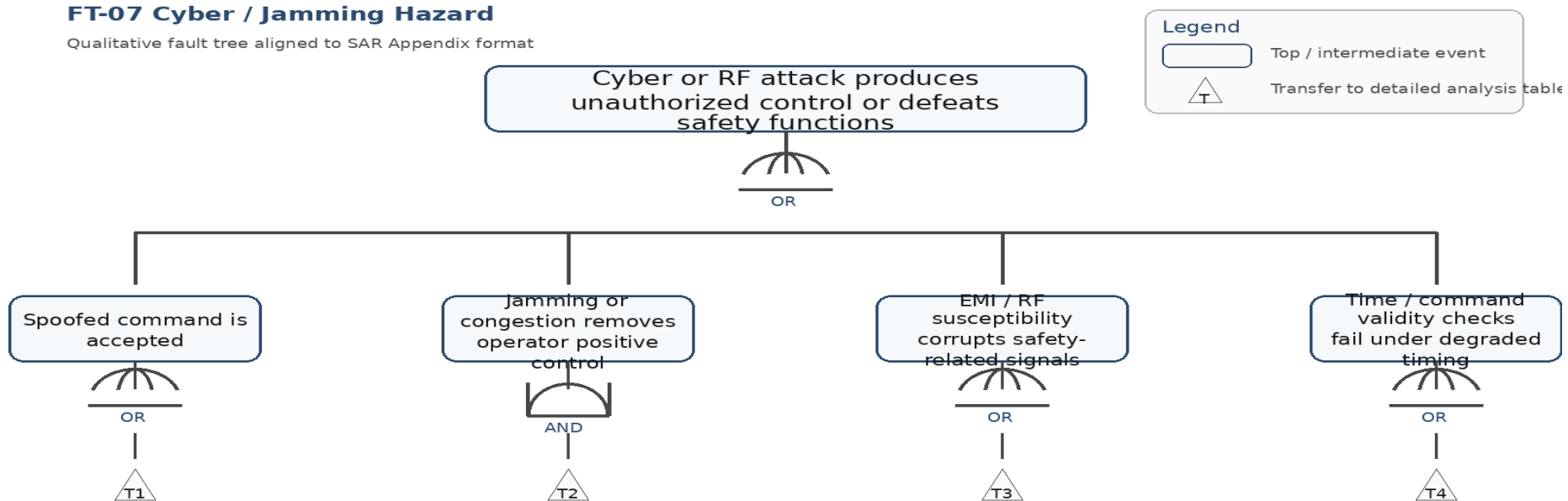
FTA assessment: This tree ties thermal hazards to duty-cycle control, barrel temperature sensing, BMS controls, and stall shutdown. The top event is prevented by both limiting heat generation and reliably inhibiting continued firing. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

10.FT-07 Cyber / Jamming Hazard

PHA hazard	Cyber/jamming	Related SHA hazard IDs	H-8, H-9, H-29, H-31, H-40
Top event	Cyber or RF attack produces unauthorized control or defeats safety functions		
Tree logic	OR	Primary appendix figure	Figure FT-07
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	This tree bridges safety and RMF concerns. It shows how authentication, anti-replay, EMI hardening, and deterministic fail-safe transitions form the cyber-safety barrier set.		

FT-07 Cyber / Jamming Hazard

Qualitative fault tree aligned to SAR Appendix format



FT-07 Cyber / Jamming Hazard

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-9	Spoofed / unauthorized command is accepted	H-9: Weak auth; key compromise; spoofed packets	H-9: Strong encryption + mutual auth; rolling keys; whitelist device IDs; command signing; anti-repl...	H-9: Pen-test; protocol tests; replay/spoof tests; code review
T2	H-8, H-40	Jamming or congestion removes control and SAFE transition fails	H-8: RF link drop/jam; integration fault H-40: Camera failure; link congestion	H-8: Heartbeat supervision; automatic DISARM on link loss; local timer to safe; operator re-auth req... H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l...	H-8: Comm-loss tests (RF off/jam); timing verification; flight test H-40: Video loss tests; comm congestion tests; inhibit demo
T3	H-29, H-4	EMI / RF susceptibility corrupts control or inhibit lines	H-29: RF emitters on DRONE; payload interference H-4: EMI coupling into trigger line; ESD	H-29: EMI shielding; filtering; separation; grounding; software robustness H-4: EMI/ESD filtering; shielded cables; debounce + command authentication; redundant inhibit; groun...	H-29: EMI tests; on-platform integration test H-4: EMI/ESD tests (MIL-STD-461 as applicable); bench susceptibility tests
T4	H-31, H-9	Time validity / anti-replay checks defeated by drift or sync loss	H-31: Time drift; GPS time loss H-9: Weak auth; key compromise; spoofed packets	H-31: Monotonic timers; sanity checks; disable time-based gating on invalid time; alarm H-9: Strong encryption + mutual auth; rolling keys; whitelist device IDs; command signing; anti-repl...	H-31: Time-drift tests; GPS-loss tests H-9: Pen-test; protocol tests; replay/spoof tests; code review

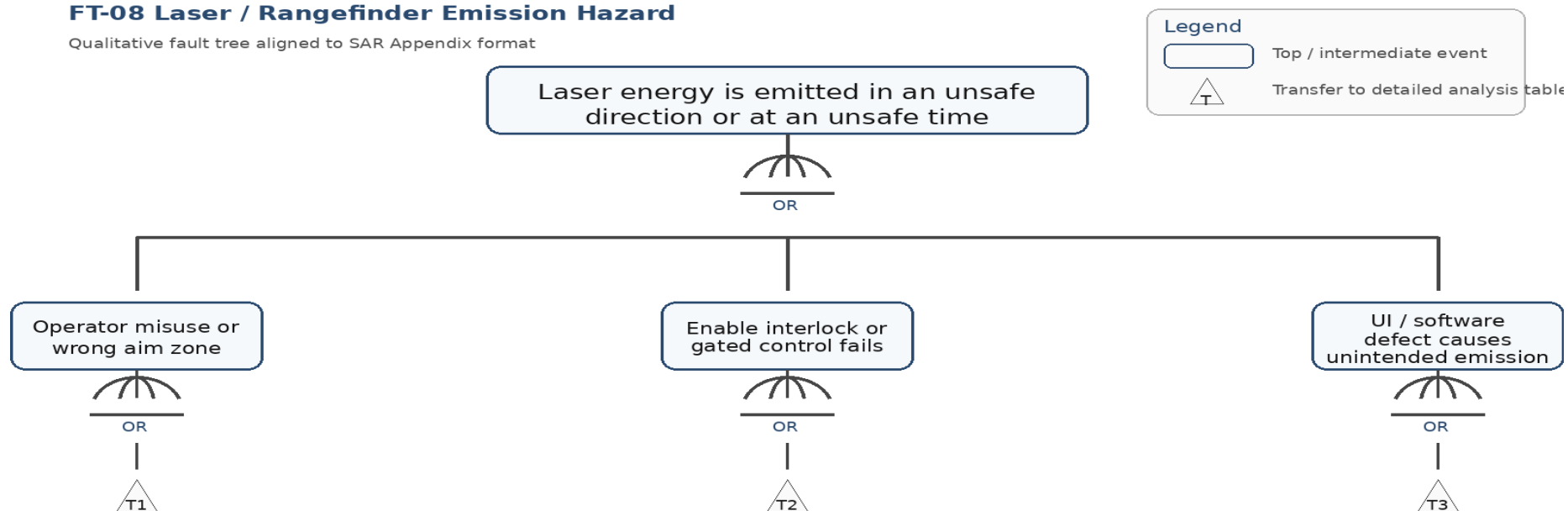
FTA assessment: This tree bridges safety and RMF concerns. It shows how authentication, anti-replay, EMI hardening, and deterministic fail-safe transitions form the cyber-safety barrier set. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

11.FT-08 Laser / Rangefinder Emission Hazard

PHA hazard	Laser hazard	Related SHA hazard IDs	H-21, H-22
Top event	Laser energy is emitted in an unsafe direction or at an unsafe time		
Tree logic	OR	Primary appendix figure	Figure FT-08
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	Although not part of the firing chain, the optical emission hazards remain Cat II/Cat I safety concerns and therefore merit an explicit appendix tree.		

FT-08 Laser / Rangefinder Emission Hazard

Qualitative fault tree aligned to SAR Appendix format



בלמ"ס – מסווג מסחרית

FT-08 Laser / Rangefinder Emission Hazard

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-21	Operator uses laser in unsafe zone or without proper warning	H-21: Operator misuse; interlock failure	H-21: Laser class management; gated enable; interlock with aim zone; warning indicators; SOP; auto ti...	H-21: Laser safety test; interlock verification; operator training records
T2	H-21, H-22	Gated enable, interlock, or timeout fails	H-21: Operator misuse; interlock failure H-22: UI bug; stuck enable	H-21: Laser class management; gated enable; interlock with aim zone; warning indicators; SOP; auto ti... H-22: Positive enable + indicator; time-out; separate control path; BIT	H-21: Laser safety test; interlock verification; operator training records H-22: Functional tests; UI regression; emission enable tests
T3	H-22, H-32	UI / software bug leaves emission enabled when not intended	H-22: UI bug; stuck enable H-32: UI design; accidental click	H-22: Positive enable + indicator; time-out; separate control path; BIT H-32: Guarded mode switch; confirm dialog; logging; audible/visual annunciation	H-22: Functional tests; UI regression; emission enable tests H-32: Human factors test; UI validation

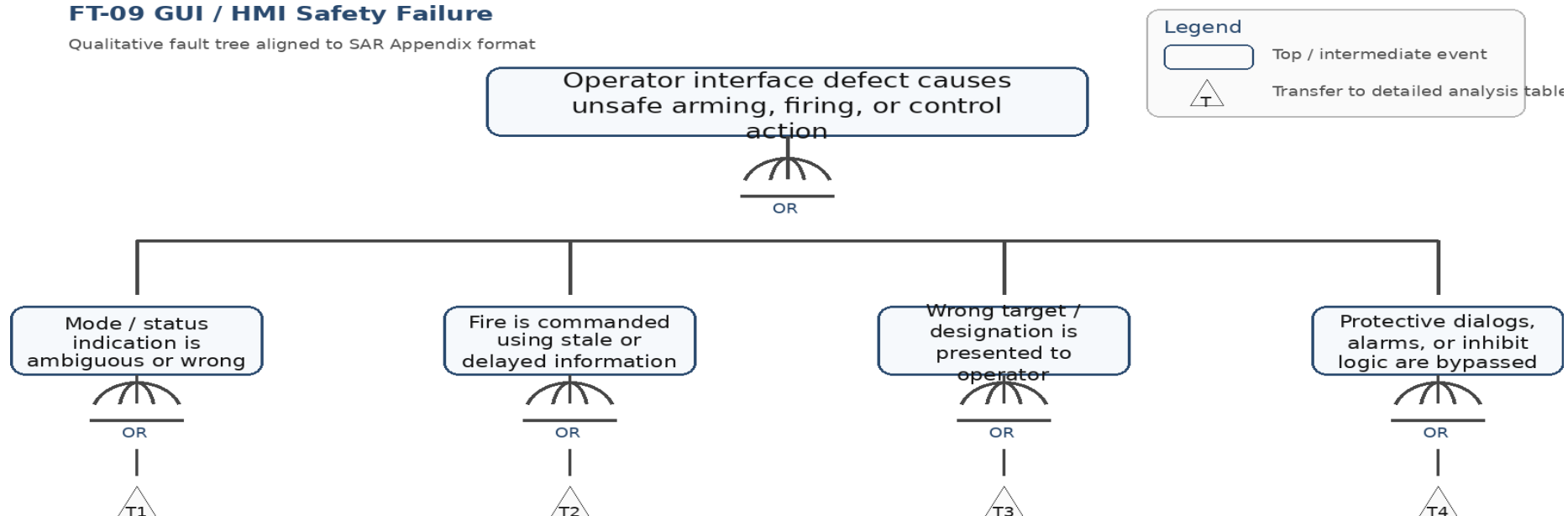
FTA assessment: Although not part of the firing chain, the optical emission hazards remain Cat II/Cat I safety concerns and therefore merit an explicit appendix tree. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

12.FT-09 GUI / HMI Safety Failure

PHA hazard	GUI failure	Related SHA hazard IDs	H-3, H-10, H-32, H-39, H-40
Top event	Operator interface defect causes unsafe arming, firing, or control action		
Tree logic	OR	Primary appendix figure	Figure FT-09
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	This tree supports MIL-STD-1472 style HFE safety reasoning. It highlights that UI clarity is not convenience; it is a primary safety control in a HITL weapon architecture.		

FT-09 GUI / HMI Safety Failure

Qualitative fault tree aligned to SAR Appendix format



FT-09 GUI / HMI Safety Failure

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-3, H-32	Ambiguous mode state; accidental mode change; poor annunciation	H-3: Mode confusion; accidental trigger; UI ambiguity H-32: UI design; accidental click	H-3: Two-step FIRE (arm + confirm); guarded control; clear UI state; required positive ID & authoriz... H-32: Guarded mode switch; confirm dialog; logging; audible/visual annunciation	H-3: Human factors review; UI simulation; operator eval; procedural compliance drills H-32: Human factors test; UI validation
T2	H-10, H-40	Latency, dropped packets, or video loss degrade operator picture	H-10: High latency; buffering; GUI lag H-40: Camera failure; link congestion	H-10: Time-stamped commands; aim-valid window; require stable track + operator confirm; inhibit if la... H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l...	H-10: Network emulation tests; end-to-end latency tests; firing drills H-40: Video loss tests; comm congestion tests; inhibit demo
T3	H-39, H-23	Wrong datalink target / UI mix-up shown to operator	H-39: Wrong datalink target; UI mix-up H-23: EO/IR freeze; stale frames	H-39: HITL positive ID; ROE prompts; target labeling; confirm-before-fire; logs H-23: Stale-frame detection; sensor health; overlay warnings; inhibit auto-track on stale; operator c...	H-39: Operational scenario tests; training; UI validation H-23: Sensor fault injection; frame-freeze tests; field demo
T4	H-3, H-10, H-40	Confirm dialogs, alarms, or inhibit thresholds are bypassed	H-3: Mode confusion; accidental trigger; UI ambiguity H-10: High latency; buffering; GUI lag H-40: Camera failure; link congestion	H-3: Two-step FIRE (arm + confirm); guarded control; clear UI state; required positive ID & authoriz... H-10: Time-stamped commands; aim-valid window; require stable track + operator confirm; inhibit if la... H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l...	H-3: Human factors review; UI simulation; operator eval; procedural compliance drills H-10: Network emulation tests; end-to-end latency tests; firing drills H-40: Video loss tests; comm congestion tests; inhibit demo

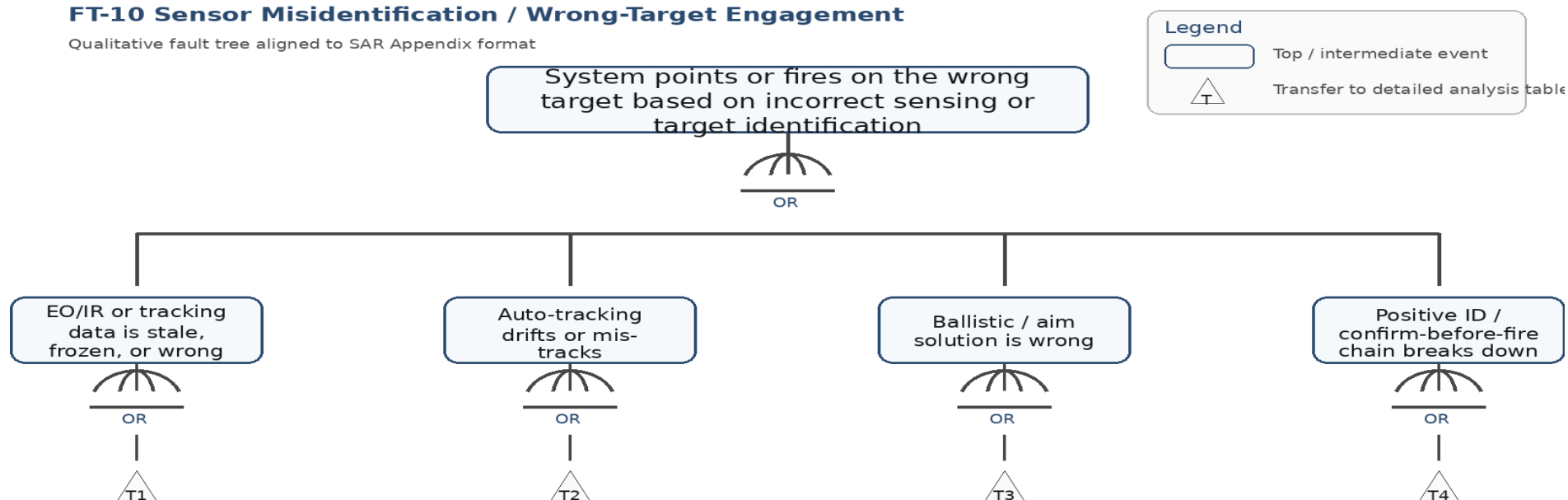
FTA assessment: This tree supports MIL-STD-1472 style HFE safety reasoning. It highlights that UI clarity is not convenience; it is a primary safety control in a HITL weapon architecture. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

13.FT-10 Sensor Misidentification / Wrong-Target Engagement

PHA hazard	Sensor misID	Related SHA hazard IDs	H-23, H-24, H-25, H-39, H-40
Top event	System points or fires on the wrong target based on incorrect sensing or target identification		
Tree logic	OR	Primary appendix figure	Figure FT-10
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	This tree is central to the authority-signoff hazard for wrong-target engagement. The analysis shows why HITL confirmation must remain independent of automation confidence.		

FT-10 Sensor Misidentification / Wrong-Target Engagement

Qualitative fault tree aligned to SAR Appendix format



בלמ"ס – מסווג מסחרית

FT-10 Sensor Misidentification / Wrong-Target Engagement

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-23, H-40	EO/IR freeze, stale frame, or bad aim indication	H-23: EO/IR freeze; stale frames H-40: Camera failure; link congestion	H-23: Stale-frame detection; sensor health; overlay warnings; inhibit auto-track on stale; operator c... H-40: Automatic DISARM if video lost beyond threshold; fallback sensor; alarm; inhibit FIRE without l...	H-23: Sensor fault injection; frame-freeze tests; field demo H-40: Video loss tests; comm congestion tests; inhibit demo
T2	H-24	Auto-tracking drifts / mis-tracks under clutter	H-24: Algorithm error; clutter	H-24: HITL: require explicit operator confirm before FIRE; track confidence gating; easy drop-track c...	H-24: Tracking evaluation; scenario tests; operator-in-the-loop trials
T3	H-25	Ballistic solution wrong due to profile, zero, or wind input	H-25: Bad zero; wrong ammo profile; wind input	H-25: Zeroing procedure; ammo profile control; plausibility checks; display assumptions; inhibit if n...	H-25: Range zero tests; ballistic regression; configuration tests
T4	H-39, H-24	HITL positive ID / ROE / confirm-before-fire chain breaks down	H-39: Wrong datalink target; UI mix-up H-24: Algorithm error; clutter	H-39: HITL positive ID; ROE prompts; target labeling; confirm-before-fire; logs H-24: HITL: require explicit operator confirm before FIRE; track confidence gating; easy drop-track c...	H-39: Operational scenario tests; training; UI validation H-24: Tracking evaluation; scenario tests; operator-in-the-loop trials

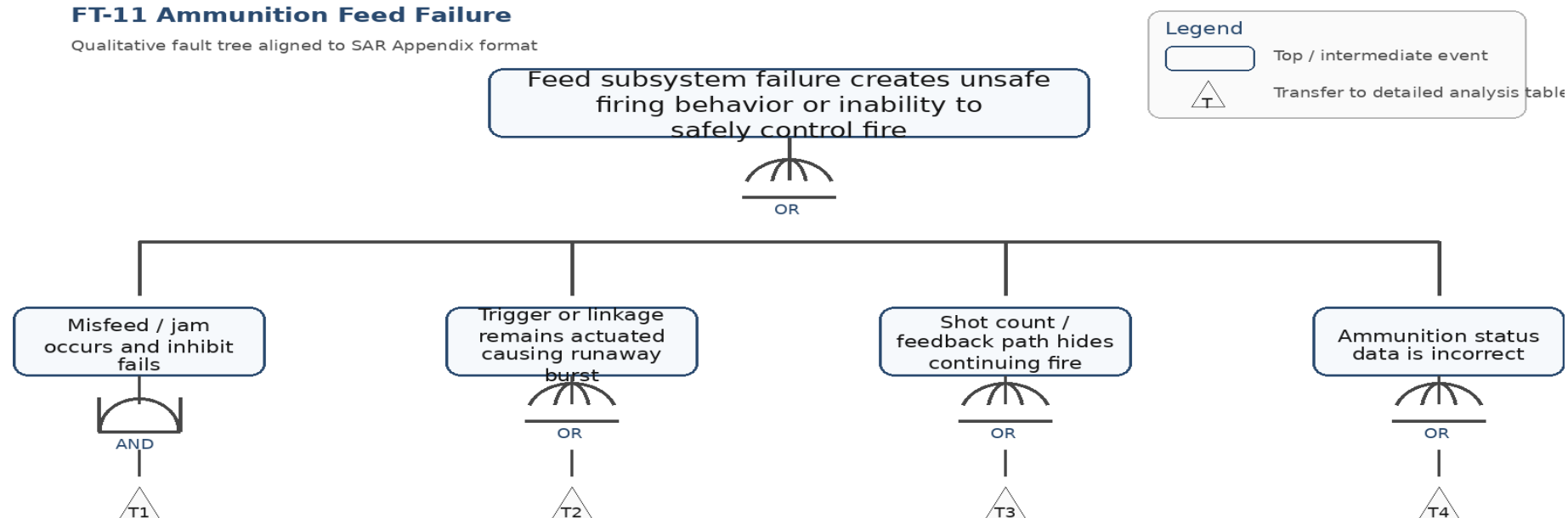
FTA assessment: This tree is central to the authority-signoff hazard for wrong-target engagement. The analysis shows why HITL confirmation must remain independent of automation confidence. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

14.FT-11 Ammunition Feed Failure

PHA hazard	Feed failure	Related SHA hazard IDs	H-15, H-16, H-26, H-37
Top event	Feed subsystem failure creates unsafe firing behavior or inability to safely control fire		
Tree logic	OR	Primary appendix figure	Figure FT-11
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	The feed path is safety-critical because jams and hidden continuing fire can rapidly escalate into thermal, control, and unauthorized-fire hazards.		

FT-11 Ammunition Feed Failure

Qualitative fault tree aligned to SAR Appendix format



FT-11 Ammunition Feed Failure

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-15	Feed jam occurs and trigger inhibit or jam-clear discipline fails	H-15: Feed chute damage; belt misalignment	H-15: Jam detection; immediate trigger inhibit; jam-clear procedure with DISARM required; robust feed...	H-15: Jam tests; maintainability demo; induced misfeed trials
T2	H-16	Trigger stuck or linkage jam creates continuing burst	H-16: Mechanical trigger linkage jam; solenoid stuck	H-16: Burst limiter; cut-off timer; redundant inhibit; monitored firing feedback (shot counter) to st...	H-16: Burst-limit test; stuck-solenoid fault test; live-fire demo
T3	H-37	Shot counter / feedback sensor fails and abnormal firing is hidden	H-37: Sensor disconnected	H-37: Independent burst limiter; redundant detection (acoustic/accel optional); health monitoring	H-37: Sensor disconnect tests; burst limiter demo
T4	H-26	Ammo count or belt state is wrong and operator decision is degraded	H-26: Sensor error; software counter drift	H-26: Dual method (shot count + belt sensor if available); reconcile on reload; warning on low confid...	H-26: Counter validation; reload scenario tests

FTA assessment: The feed path is safety-critical because jams and hidden continuing fire can rapidly escalate into thermal, control, and unauthorized-fire hazards. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.

15.FT-12 Structural / Retention Failure

PHA hazard	Structural failure	Related SHA hazard IDs	H-13, H-27, H-28, H-35
Top event	Structural or retention failure causes payload release, unsafe pointing, or loss of vehicle control		
Tree logic	OR	Primary appendix figure	Figure FT-12
Architecture focus	Mapped to relevant command, actuation, sensing, power, and procedural barriers		
Analysis note	This tree addresses structural integrity, retention, and routing-induced loads as a single appended safety case because all can lead to catastrophic drop or instability outcomes.		

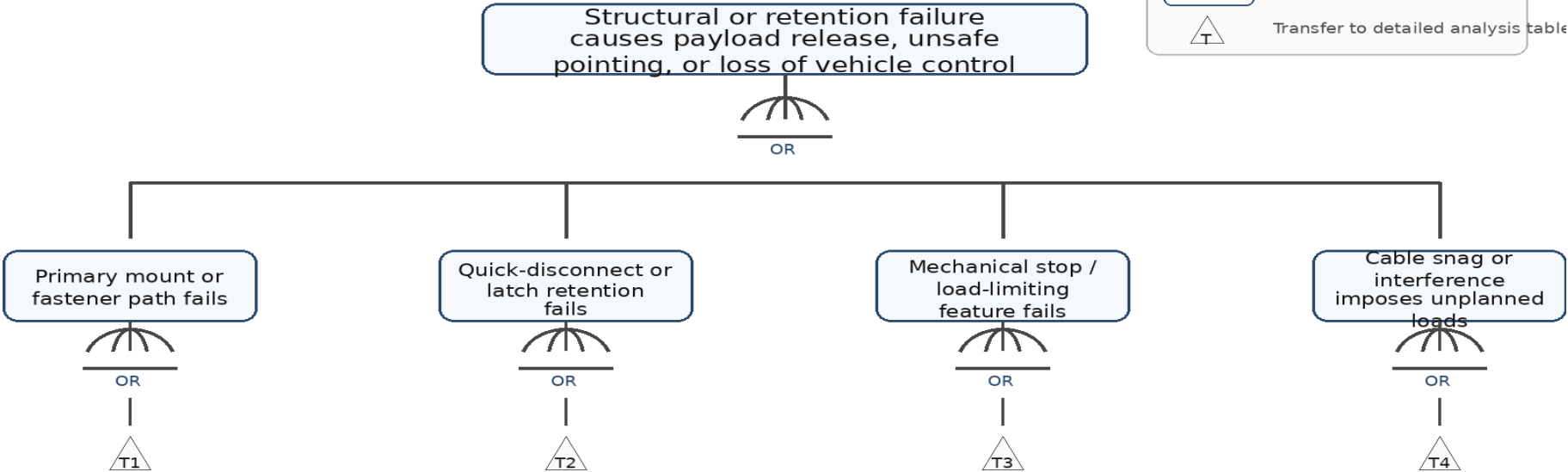
FT-12 Structural / Retention Failure

Qualitative fault tree aligned to SAR Appendix format

Legend

Top / intermediate event

Transfer to detailed analysis table



FT-12 Structural / Retention Failure

Transfer node	Related hazard ID(s)	Contributing event / cause basis	Hazard-log cause statement	Mitigation(s) from SHA	Verification / evidence from SHA
T1	H-13	Fastener loosening, fatigue, or overload breaks primary mount	H-13: Fastener loosening; fatigue; overload landing	H-13: Structural analysis with safety factors; locking fasteners; secondary retention; periodic torqu...	H-13: Static load + fatigue test; vibration test; inspection
T2	H-35	Latch / quick-disconnect improperly assembled or fails in service	H-35: Latch failure; improper assembly	H-35: Positive locking + secondary latch; assembly checklist; visual indicator; torque marks	H-35: Drop/load test; assembly verification; inspection
T3	H-27	Mechanical stop fracture or wear removes protective load limit	H-27: Stop fracture; wear	H-27: Stop strength margin; redundant limit switches; current limiting; periodic inspection	H-27: End-stop overload test; inspection; motion tests
T4	H-28	Cable routing / snag imposes side load or interference on structure	H-28: Routing error; wear	H-28: Cable management; strain relief; clearance analysis; protective sleeving	H-28: Fit-check; vibration test; flight envelope inspection

FTA assessment: This tree addresses structural integrity, retention, and routing-induced loads as a single appended safety case because all can lead to catastrophic drop or instability outcomes. The transfer-node table preserves direct linkage from each branch to the exact SHA hazard IDs, mitigations, and verification methods used in the source package.